



Compaction effects on soil macropore geometry and related parameters for an arable field

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ABSTRACT

Soil compaction decreases soil pores are important for root growth as well as infiltration of water and nutrients. A study was conducted to evaluate the effects of soil compaction on macropore parameters measured using X-ray computed tomography (CT). Macropore parameters included number of pores, number of macropores (>1000 µm diam.), number of coarse mesopores (200 to 1000 µm diam.), porosity, macroporosity, coarse mesoporosity, area of largest pore, pore circularity, and fractal dimension of macroporosity. A field experiment was conducted on Mexico silt loam (fine, smectitic, mesic Vertic Epiaqualfs) with field treatments including four replicates of uniformly Compacted (C) and Non-Compacted (NC) plots arranged in a randomized complete block design. Soil cores (76.2 mm diam. by 76.2 mm long) were removed from three selected depths (0 to 10 cm, 10 to 20 cm, and 20 to 30 cm). Cores were scanned using a medical X-ray CT scanner with four scans taken in each sample at 15 mm spacing starting at 25 mm from the core surface. Images were analyzed using *Image-J* software. The C treatment was found to increase bulk density by 8% (1.34 to 1.45 g cm⁻³) and decrease saturated hydraulic conductivity by 69% (47.1 to 14.6 cm hr⁻¹). CT-measured number of pores decreased by 71%, number of macropores by 69%, and coarse mesopores by 75% with the C treatment used in the study. Compaction was also found to significantly decrease CT-measured porosity and macroporosity by 64%. Differences between treatments for the parameters were most pronounced in the upper 10 cm; differences between treatments were not significant below 20 cm. A regression equation with CT-measured macroporosity, area of largest pore and porosity explained most of the variability in saturated hydraulic conductivity ($R^2=0.79$). Efforts should be made to minimize soil compaction due to its harmful effects on soil pores and subsequent challenges for plant root growth and enhanced runoff of water and nutrients.

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1. Introduction

Concerns about soil compaction have increased due to the use of heavy equipment which compresses the soil and reduces the potential for optimal conditions for grain crop production (Soane and van Ouwerkerk, 1995; Entry et al., 1996; Abu-Hamdeh, 2003). Harmful effects of compaction on soil physical parameters as well as crop production have been reported (Motavalli et al., 2003). Compaction reduces total porosity and increases soil bulk density, as well as reduces the proportion of larger pores to smaller pores in the soil (Motavalli et al., 2003; Schaffer et al., 2008a,b). In addition, soil compaction may increase nutrient runoff (Brussard and van Faasen, 1994; Soane and van Ouwerkerk, 1995). These physical effects of soil compaction may decrease water and nutrient availability, aeration,

and plant production due to limitations in root growth (Domzal et al., 1991; Mapfumo et al., 1998; Arocena, 2000).

Some researchers have assessed whether soil compaction can be modified with applications of poultry litter (Pengthamkeerati et al., 2006). These researchers found that additions of poultry litter improved soil physical and biological properties. Compaction on claypan soils is dependent on the thickness of the overlying surface horizon above the argillic claypan horizon with shallow surface layers enhancing conditions for surface compaction (Pengthamkeerati et al., 2005). This is due in part to higher water content conditions in the surface horizon attributed to the low saturated hydraulic conductivity of the argillic horizon in claypan soils.

The study presented was conducted on claypan soils. Claypan soils occur in approximately 4 Mha in the Midwestern USA (Blanco-Canqui et al., 2002). These soils are characterized by silt loam surface horizons with an abrupt argillic horizon within 10 to 50 cm of the soil surface. This argillic horizon has a significant smectitic clay content (40 to 60%) which limits vertical water drainage and causes episodic perching of water in the surface horizon (Blanco-Canqui et al.,

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2002). These soils are often saturated after rainfall events and are often very moist in the spring during tillage and planting operations which enhances compaction.

Image analysis techniques have been used to quantify soil structure in the past (Moran et al., 1988; McBratney and Moran, 1990; McBratney et al., 1992; Moran and McBratney, 1992). These researchers used 2-dimensional images of soil profiles for the determination of several structural attributes. Moran et al. (1988) and Douglas et al. (1992) used rectangular soil samples about 15 to 20 cm wide and 20 to 30 cm long to evaluate soil structure. Douglas et al. (1992) were able to determine better soil structure in uncompacted soil treatments using these image analysis techniques. Recent advances in computed tomography (CT) may play a role in quantifying soil structural features (Rachman et al., 2005).

X-ray CT has been used in recent years as a new method for quantifying soil pores, especially macropore development, at a much higher resolution compared to prior methods (Grevers and de Jong, 1994; Francis et al., 2001; Capowiez et al., 2003). Some investigators have used dual energy X-ray CT techniques for measuring soil phases for use in assessing soil structure (Rogasik et al., 1999). Aylmore (1994) reviewed many research studies which have used CT for measurement of plant root systems as well as soil water uptake (Aylmore, 1994). Rachman et al. (2005) showed that CT-measured macropore parameters were found to have close agreement with water retention-derived measurements. Other researchers (Udawatta et al., 2006) have utilized CT measurement tools to evaluate and compare the effects of different management systems such as buffers on pore parameters. Rogasik et al. (2003) used CT to assess soil structure using CT to compare small scale compaction near earthworm burrows and also analyze the effects of tillage. Some recent studies have used 3-dimensional characterization of soil pores and macropores (Perret et al., 1999; Pierret et al., 2002).

Since soil macropores are often produced by earthworms in soils, several researchers have evaluated burrowing effects of earthworms using CT systems. Jegou et al. (2002) evaluated the effects of soil compaction on earthworm burrow systems using CT measurement systems. A CT evaluation of earthworm burrowing effects on repacked soil cores were studied by Bastardie et al. (2003). Bastardie et al. (2005) also evaluated earthworm systems using CT in natural soil cores from pastures. Schrader et al. (2007) using CT evaluated the bulk density on the edge of earthworm burrows; they found these areas had 11% higher bulk density and that soil compaction decreased the diameter of earthworm burrows.

The objective of the study was to evaluate the effects of soil compaction on macropore geometry and related parameters measured using X-ray computed tomography (CT). Macropore parameters include number of pores, number of macropores ($>1000\ \mu\text{m}$ diam.), number of coarse mesopores (200 to $1000\ \mu\text{m}$ diam.), porosity, macroporosity, coarse mesoporosity, area of largest pore, pore circularity, and fractal dimension of macroporosity.

2. Materials and methods

2.1. Experimental site

A field experiment was conducted at the Bradford Research and Extension Center near Columbia, Missouri ($38^\circ 53' \text{N}$ and $92^\circ 13' \text{W}$). The site was planted to corn (*Zea mays* L.) during the two years prior to the study. Soil at the site was in the Central Claypan Area MLRA (major land resource area) and classified as Mexico silt loam (fine, smectitic, mesic Vertic Epiaqualfs) which formed in loess over glacial till. The depth to the claypan at the site varied from 25 to 30 cm from the soil surface. Soil physical and chemical properties for the site are illustrated in Table 1.

The area selected for the study was chisel plowed to a depth of 20 cm during the fall and then disked twice in the spring to a depth of

Table 1

Soil physical and chemical properties as a function of depth for the study site.

Depth (cm)	Soil texture			pH (water)	Total organic C (g kg^{-1})
	Sand (g kg^{-1})	Silt (g kg^{-1})	Clay (g kg^{-1})		
0 to 10	57 (4)	737 (15)	206 (12)	6.04 (0.10)	15.7 (0.8)
10 to 20	50 (2)	725 (19)	225 (21)	6.05 (0.16)	14.7 (0.7)
20 to 30	36 (6)	591 (83)	373 (93)	5.75 (0.33)	13.5 (0.6)

Averages are shown with standard error of the means in parentheses.

15 cm prior to imposing treatments. The experimental area was treated as fallow (no plants grown) with plot sizes of 3.0 m wide $\times$ 6.1 m long. The plots were treated with periodic applications of glyphosate for weed control. Wheel traffic for corn planting and herbicide application was imposed on the fallow plots during the experiment; no corn planting occurred but traffic appropriate to planting was imposed on the plots.

Field treatments included four replicates of uniformly compacted and non-compacted plots arranged in a randomized complete block design. These treatments are referred to as Compacted (C) and Non-Compacted (NC) treatments. The surface compaction treatment consisted of two passes with a tractor pulling a water wagon. The tractor (Model 504 International Harvester) had an axle load of 2.5 Mg and the water wagon (1890 L water) had an axle load of 2.9 Mg. Front and rear tires, under 221 kPa inflation pressure, on the water wagon were 17.8 cm wide. Vertical stress on the soil surface was estimated to be approximately 500 to 1000 kPa. The water wagon made two passes over the soil so that uniform wheel track coverage occurred over the whole plot areas for the C treatment; this compaction treatment was only for the C treatment. Effort was made so that in one pass wheels of the water wagon covered all of the soil area; an additional pass with the wheels of the water wagon was conducted with all of the soil area compacted a second time.

2.2. Sample collection

Intact soil cores were removed in the early summer from the C and NC treatments to determine compaction effects on CT-measured macropore properties. The dimensions of the sampling Plexiglas rings were 76.2 mm long and 76.2 mm diam. with a 3.2 mm thick wall. Twenty-four intact soil cores were collected from three soil depths (0 to 10 cm, 10 to 20 cm and 20 to 30 cm) per treatment with four replications per treatment. These 24 samples were removed from the non-trafficked inter-row position of the NC and C treatment plots. It is noted that the C treatment plots were completely trafficked to create the compaction treatment. The equivalent location of the non-trafficked inter-row position within the NC treatment plots was also sampled in the C treatment plots for a consistent comparison.

In addition to these 24 samples, an additional eight core samples were taken from the 0 to 10 cm depth in the NC treatment plots to assess the effects of traffic (only within the NC treatment). Four samples were removed from the trafficked inter-row position and four from the non-trafficked inter-row position within the NC treatment, corresponding to the four replicates. These treatments are referred to as Trafficked and Non-Trafficked (within the NC main treatment).

Core samples were trimmed, labeled, and sealed in plastic bags, and then transported to the laboratory. Samples were stored at 4°C until measurements were taken.

2.3. Scanning and image analysis

Soil core samples were removed from cool storage and saturated with a dilute salt solution (CaCl_2 ; $6.24\ \text{g L}^{-1}$ and MgCl_2 ; $1.49\ \text{g L}^{-1}$) to maintain soil structure. After 48 h, sample weights were recorded and samples were then drained at 5.0 kPa tension for 24 h using a glass-bead tension table. This procedure removed water from pores $>60\ \mu\text{m}$

equivalent cylindrical diameter which enhanced image contrast between air-filled pores and moist soil solids.

The cores were then scanned using a medical GE Genesis-Zeus X-ray CT scanner set at a peak voltage of 120 keV and a current of 100 mA to acquire CT scan images. Soil cores were positioned horizontally on the scanner stage so that the X-ray beam was perpendicular to the longitudinal axis. Four images were acquired from each core at the following scan depths from the core surface: 2.5, 4.0, 5.5, and 7.0 cm (actual soil depths were 3.0, 4.5, 6.0 and 7.5 cm from 0 cm, 10 cm, or 20 cm depths). The pixel resolution was 0.19 by 0.19 mm and the slice thickness was 1.0 mm, producing a voxel size of 0.036 mm³. A total of 128 images were analyzed in this study.

Images were analyzed using the *Image-J* ver. 1.27 software (Rasband, 2002) to examine the treatment effects on pore characteristics. The *Region of Interest* (ROI) tool was used to exclude voids near the core walls and to minimize the effects of beam hardening. Pores were partitioned from solids using the *Threshold* tool after converting the image into an 8-bit grayscale image. Conversion of the image included all of the original data; known phantoms on the sample were used to validate the conversion. A value of 60 (range is 0 to 255) was selected as the threshold value selected for analysis of the images; this value was selected using optimization with known phantoms which included air-filled tubes. Values lower than the threshold value were identified as air-filled pores and values greater than threshold value were identified as non-pore. Statistics of individual pores were computed with the *Analyze Particles* tool. Macropore and mesopore characteristics analyzed in the scans included the number of pores, number of macropores, number of coarse mesopores, porosity (macroporosity plus coarse mesoporosity), macroporosity (>1000-μm diam.), and coarse mesoporosity (200- to 1000-μm diam.). Macroporosity and mesoporosity at each scan depth were calculated from the total area of all macropores and mesopores isolated in the image at a given depth divided by the cross sectional area (4200 mm²) of the selected region from the soil core image (Fig. 1). In addition, the area of the largest pore, pore circularity, and fractal dimension of macroporosity were determined. Dividing the product of the pore area and 4π by the pore perimeter squared was the method used to estimate circularity of macropores (Tuller et al., 1999). The circularity was obtained by the following equation:

$$C = \frac{4\pi(A_p)}{(P_p)^2} \quad (1)$$

where A_p is the pore area and P_p is the pore perimeter.

To estimate fractal dimension of macroporosity, a rectangular region of 50 by 50 mm area was used with the *Region of Interest* (ROI) tool. Pores were partitioned from solids using the *Threshold* tool after converting the image into an 8-bit grayscale image. The fractal dimension of macropores was determined with zero to 100 threshold values to better populate low porosity samples with pores. This value is slightly higher than the value of 60 used for pores; it is similar to the value used by Gantzer and Anderson (2002). The fractal dimension for all images was obtained as the slope of the log–log plot of measured box count as a function of box size.

2.4. Bulk density and saturated hydraulic conductivity

Subsequent to scanning, saturated hydraulic conductivity and bulk density were determined for the soil core samples. Saturated hydraulic conductivity was determined using the constant head method (Klute and Dirksen, 1986). Soil cores were then dried at 105 °C until constant weight was obtained (about 48 h). Bulk density was then determined as described by Blake and Hartge (1986). Saturated hydraulic conductivity values were compared with the CT-measured pore parameters. Saturated hydraulic conductivity was not measured on core samples from the trafficked and non-trafficked inter-row samples.

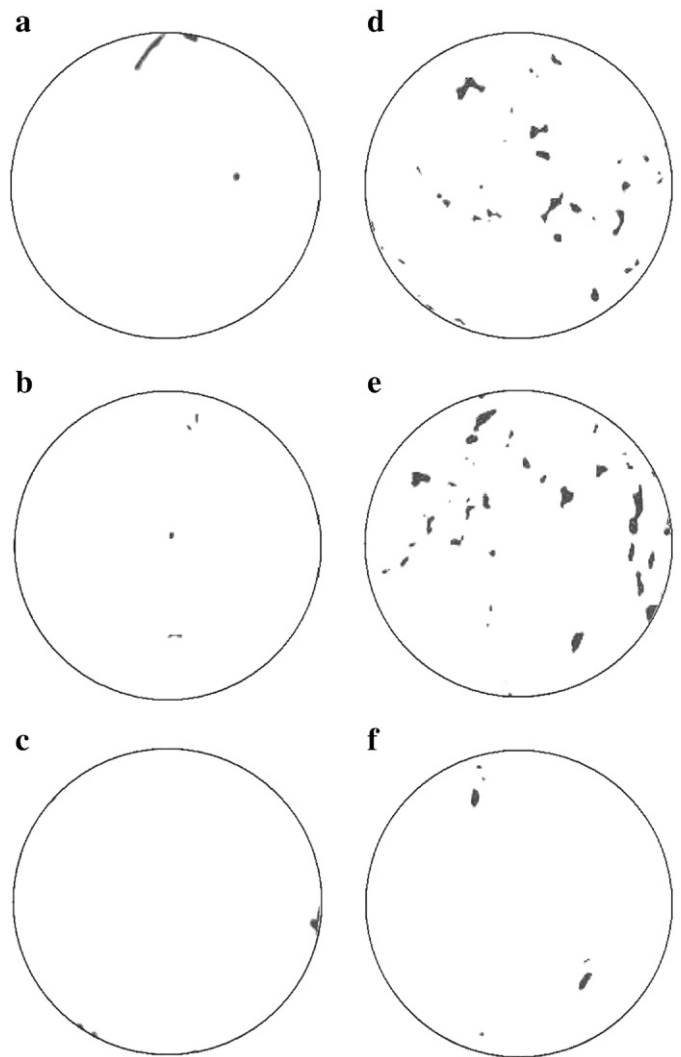


Fig. 1. Selected scan images (73 mm diameter) with pores (pores = black). (a) Compacted 0–10 cm depth, (b) Compacted 10–20 cm depth, (c) Compacted 20–30 cm depth, (d) Non-Compacted 0–10 cm depth, (e) Non-Compacted 10–20 cm depth, and (f) Non-Compacted 20–30 cm depth.

2.5. Statistical analysis

Analysis of variance (ANOVA) was conducted using the GLM procedure (SAS Institute, 1999) to evaluate the effects of compaction management and soil depth. Soil depth will be referred to as depth zones (0 to 10, 10 to 20, and 20 to 30 cm) and scan depths (scan depth below the soil surface). Differences in pore characteristics among scan depths throughout the profile were statistically compared to evaluate the effects of depth and compaction management. An estimate of the least significant difference (Duncan's LSD) between treatments at the same depth or different depths was obtained. Statistical differences were declared significant at the 0.05 level. Regression equations were developed between saturated hydraulic conductivity and the CT-measured pore parameters.

3. Results

3.1. CT-measured number of pores, macropores and coarse mesopores

CT-measured pores represent the pores estimated for each scan detected by the CT scanner. CT-measured number of pores was significantly affected by compaction ($P < 0.05$) and soil depth ($P < 0.01$, Table 2). CT-measured number of pores was 250% higher for the NC

Table 2

Average CT-measured number of pores (pores, macropores, and coarse mesopores), porosity (porosity, macroporosity, and coarse mesoporosity), area of largest pore, circularity, and fractal dimension of macropores as influenced by Compacted and Non-Compacted treatments and soil depth zones along with the analysis of variance.

	Number			Porosity			Area of largest pore (mm ²)	Circularity	Fractal dimension
	Pores	Macropores	Coarse mesopores	Porosity (m ³ m ⁻³)	Macro-porosity (m ³ m ⁻³)	Coarse mesoporosity (m ³ m ⁻³)			
<i>Treatment means</i>									
Compacted	6 ^b	4 ^b	2 ^b	0.0071 ^b	0.0069 ^b	0.0002 ^a	14.81 ^b	0.76 ^a	1.11 ^b
Non-Compacted	21 ^a	13 ^a	8 ^a	0.020 ^a	0.019 ^a	0.0006 ^a	22.28 ^a	0.79 ^a	1.17 ^a
<i>Depth zone means</i>									
0- to 10-cm	22 ^a	14 ^a	8 ^a	0.028 ^a	0.027 ^a	0.0006 ^{ab}	31.88 ^a	0.76 ^a	1.25 ^a
10- to 20-cm	11 ^b	7 ^b	4 ^b	0.0083 ^b	0.0080 ^b	0.0003 ^{bc}	11.10 ^b	0.81 ^a	1.11 ^b
20- to 30-cm	7 ^b	4 ^b	3 ^b	0.0045 ^b	0.0043 ^b	0.0002 ^c	12.66 ^b	0.75 ^a	1.09 ^b
<i>Analysis of variance, P>F</i>									
Treatment	0.02	<0.01	0.05	0.02	0.02	0.052	0.04	0.11	0.02
Depth	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0.13	<0.01
Treatment by depth	<0.01	<0.01	<0.01	<0.01	<0.01	0.03	0.49	0.10	0.47

The ANOVA table represents significance levels among treatments and depths for the measured parameters. Within columns, values followed by the same letter for the treatments or the depths are not significantly different at the 0.05 probability level.

soil compared to C soil (Table 2). Number of pores decreased as a function of depth with the third depth zone (20 to 30 cm) having 68% fewer pores compared to the surface depth zone (0 to 10 cm, Table 2). Significant interactions between compaction treatment and depth were observed ($P<0.01$). This effect is shown in Fig. 2a with the number of pores above 30 for the NC treatment in the first three scan depths compared to 10 or below for the last four scan depths. This is in contrast to the number of pores which were at 10 or below for all scan depths with the C treatment.

The number of CT-measured macropores was also significantly affected by compaction treatment ($P<0.01$) and soil depth ($P<0.01$, Table 2). Number of macropores was three times higher in the NC treatment compared to the C treatment. These pores decreased as a function of depth zone and were more than three times lower at the 20 to 30 cm depth compared to the surface. Interactions between treatment and depth were observed ($P<0.01$, Table 2). Macropore numbers in the first three scan depths were above 20 and in the last four scan depths below 10 for the non-compacted soil (Fig. 2b). This is in contrast to the macropore numbers below 10 for all scan depths with the C treatment.

Coarse mesopore numbers were significantly reduced by compaction and increasing soil depth ($P<0.01$, Table 2). CT-measured number of coarse mesopores decreased by 62% from the first to third depth increment, and by 75% with the compaction treatment (Fig. 2c). Interactions between treatment and soil depth also occurred for the number of CT-measured coarse mesopores ($P<0.01$). Values were above 10 for the NC treatment and below five for the C treatment within the first three scan depths (Fig. 2c). However, there were no differences below about the 25 cm depth.

3.2. CT-measured porosity, macroporosity and coarse mesoporosity

CT-measured porosity includes pores detected by the CT scanner, macroporosity plus mesoporosity. CT-measured porosity was significantly affected by the compaction treatments ($P<0.05$) and soil depth ($P<0.01$). The C treatment lowered porosity by 65% compared to the NC treatment (Table 2). Porosity decreased with soil depth by 84% from the first (0 to 10 cm) to third depth zone (20 to 30 cm). Significant ($P<0.01$) interactions occurred between treatments and soil depth (Table 2). Differences existed in porosity between compaction treatments for the first three scan depths (Fig. 3a), but no differences occurred below 20 cm.

Macroporosity assessed by CT was also affected by the compaction treatments ($P<0.05$) and soil depth ($P<0.01$, Table 2). The C treatment decreased macroporosity by 64%. As with porosity, macroporosity

decreased by 84% from the first to third depth zone (Table 2). Compaction treatment by soil depth interactions occurred due to a greater decrease in macroporosity with scan depth for the NC treatment compared to the C treatment (Fig. 3b).

CT-measured coarse mesoporosity values were significantly decreased by compaction ($P=0.052$) and increasing soil depth ($P<0.01$). Interactions between treatment and soil depth for coarse mesoporosity occurred ($P<0.05$) similarly to porosity and macroporosity. Compaction was found to decrease CT-measured coarse mesoporosity by 67% with the compaction treatment and by 67% from first to third depth zone (Table 2). Some differences between treatments occurred in coarse mesoporosity above 7.0 cm, but there were no differences below 20 cm (Fig. 3c).

3.3. Additional CT-measured pore parameters

Fractal dimension of macroporosity and area of largest pore were significantly affected by compaction treatment ($P<0.05$) and soil depth ($P<0.01$, Table 2). However, circularity of CT-measured pores was not affected by treatment or soil depth. Circularity of pores is a parameter which indicates how circular pores are within the scan images; more circular pores have a circularity near 1.0. Fractal dimension was 5% higher for the NC versus C soil. In addition, the fractal dimension of macropores decreased with increasing soil depth. Higher values occurred near the surface since the pores were more space filling compared to macropores at depths greater than 10 cm.

3.4. Bulk density and saturated hydraulic conductivity

The influence of soil compaction treatments on bulk density is shown in Fig. 4. Compaction treatments ($P<0.05$) and soil depth ($P<0.01$) significantly affected soil density (Table 3). Averaged over soil depth, compaction increased bulk density by 8%. The C treatment resulted in an increase in bulk density of 12% at the 0 to 10 cm depth zone and 15% at the 10 to 20 cm depth zone, but there was no change in the 20 to 30 cm depth zone (Fig. 4).

In the case of saturated hydraulic conductivity (K_{sat}), values for the C treatment were 74% lower compared to the NC treatment for the 0 to 10 cm depth (Fig. 4). Differences between treatments were not significant for lower depths. Conductivity values decreased 90% from the 5 to 15 cm depth (Fig. 4). Lower hydraulic conductivity values for the C treatment compared to the NC treatment are attributed to the preferential removal of macropores.

Regression relationships were developed between the CT-measured pore parameters and saturated hydraulic conductivity. The best

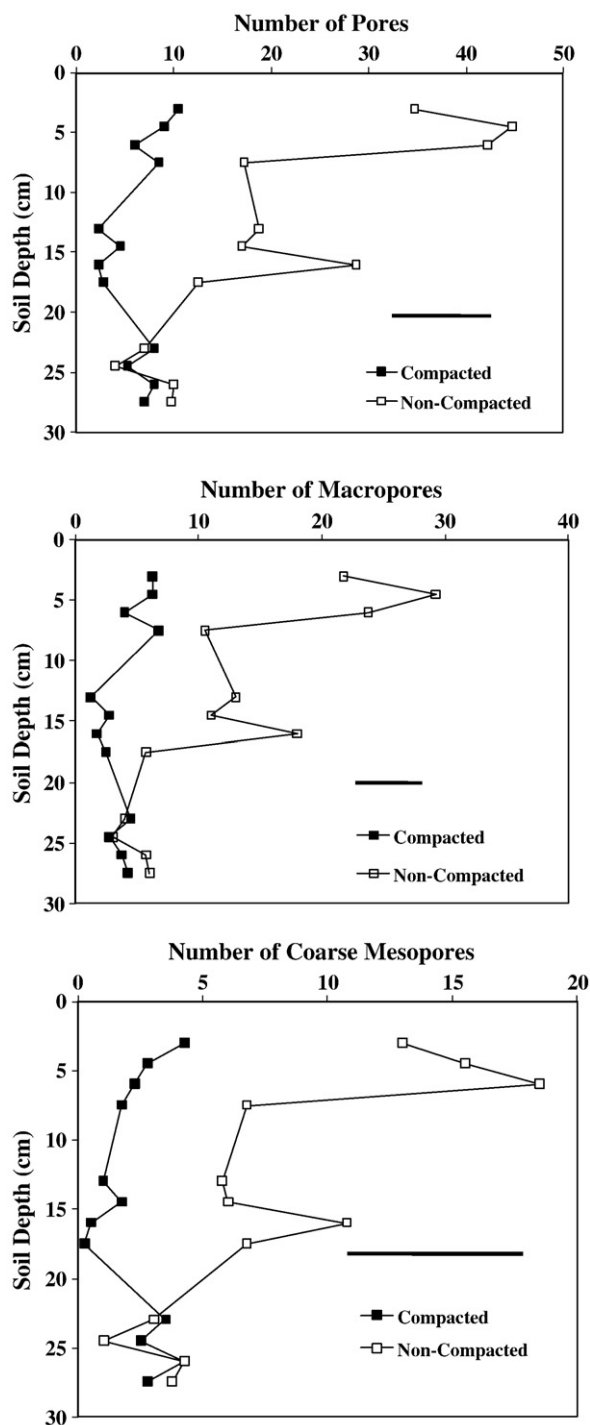


Fig. 2. CT-measured number of pores, number of macropores, and number of coarse mesopores with Compacted (closed box) and Non-Compacted (open box) as a function of soil depth. The CT-measured numbers are within a 4200 mm² scan area. Bars are LSD (0.05) values.

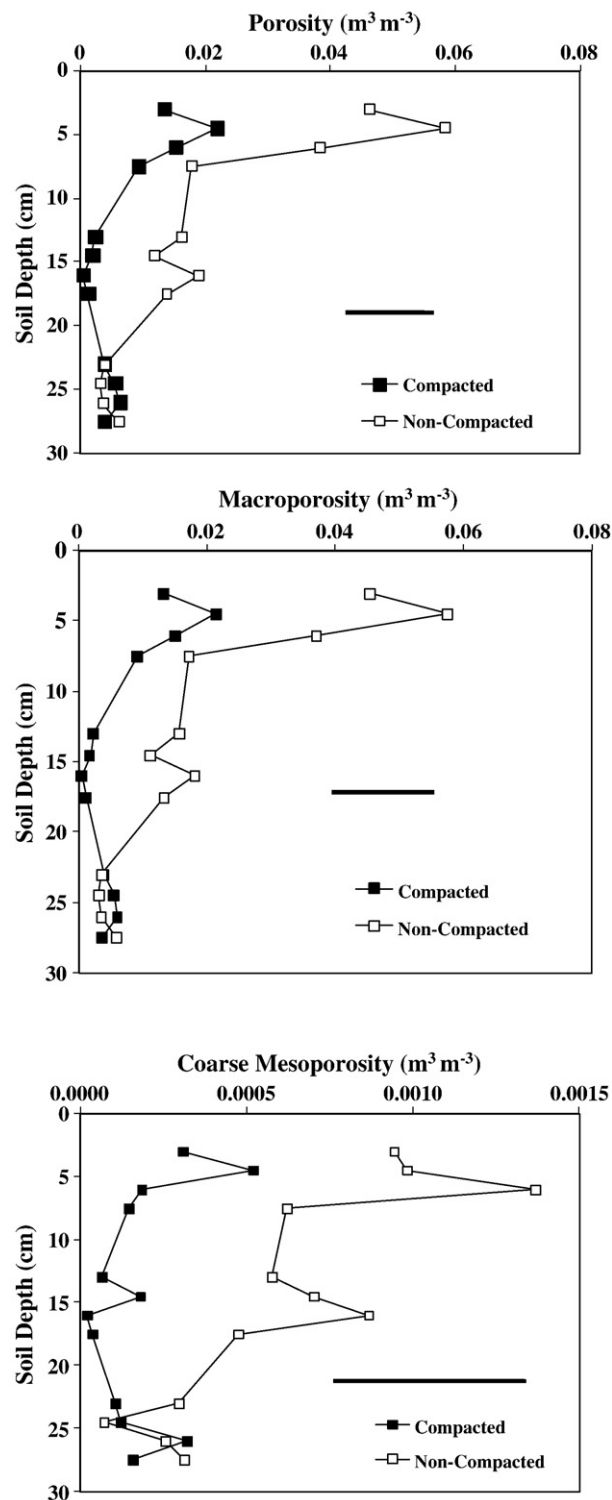


Fig. 3. CT-measured porosity, macroporosity and coarse mesoporosity with Compacted (closed box) and Non-Compacted (open box) treatments as a function of soil depth. Bars are LSD (0.05) values.

one-, two-, and three-parameter equations are shown in Table 4. Results indicate that CT-measured macroporosity best explained variations in K_{sat} for a single parameter ($r^2=0.62$). CT-measured porosity was also positively and highly correlated with K_{sat} ($r=0.787$) as were CT-measured number of pores ($r=0.744$), number of coarse mesopores ($r=0.739$), and number of macropores ($r=0.726$). Area of largest pore and CT-measured porosity were the best three-parameter model for K_{sat} ($R^2=0.79$). All parameters in the three-

parameter regression were significant at the $P<0.05$ level. The four-parameter model (area of largest pore, porosity, coarse mesoporosity, and number of coarse mesopores) increased the coefficient of determination slightly ($r^2=0.82$); however two of the parameters were not significant at the $P=0.05$ level (intercept and number of coarse mesopores).

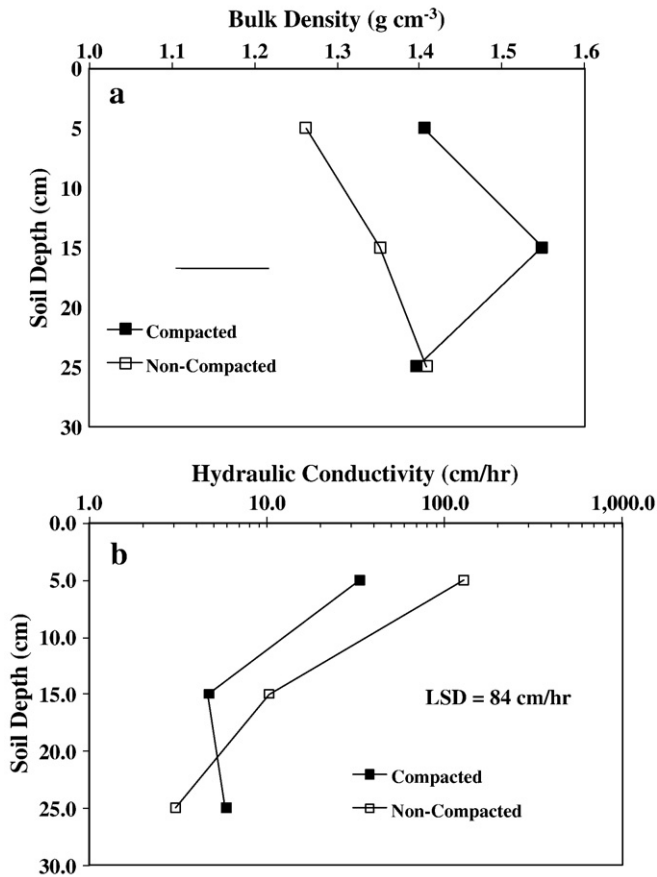


Fig. 4. Mean bulk density (a) and saturated hydraulic conductivity (b) values for the Compacted (closed box) and Non-Compacted (open box) treatments as a function of soil depth. Bar is LSD (0.05) value for bulk density; the value for conductivity is indicated on the graph due to log-scale.

3.5. Trafficked versus non-trafficked inter-rows

The number of CT-measured pores was significantly different between Trafficked soil and Non-Trafficked soil within the NC treatment. CT-measured number of pores decreased by 71% with the Trafficked treatment, and 45% from the first to the fourth scan depth (Table 5). The number of macropores determined using CT analysis decreased by 73% with the Trafficked treatment, and 50%

Table 4
Regression relationships between saturated hydraulic conductivity and CT-measured pore parameters.

Regression relationship	Coefficient of determination
<i>One parameter</i>	
$K_{sat} = -4.45 + 2695 * \text{macroporosity}$	0.62
<i>Two parameters</i>	
$K_{sat} = 11.96 + 4619 * \text{macroporosity} - 2.24 * \text{area of largest pore}$	0.75
<i>Three parameters</i>	
$K_{sat} = 21.90 + 56,016 * \text{macroporosity} - 2.87 * \text{area of largest pore} - 49,743 * \text{porosity}$	0.79

from the 3 cm to 7.5 cm scan depth. CT-measured number of coarse mesopores decreased by 69% with the Trafficked treatment.

CT-measured porosity parameters were significantly lower in Trafficked inter-rows compared to Non-Trafficked inter-rows. The Trafficked treatment decreased CT-measured porosity, macroporosity, and coarse macroporosity by 61%, 61%, and 70%, respectively (Table 5). There was a slight decrease in CT-measured porosity parameters across the four scan depths. Soil bulk density was increased by 10% for the Trafficked inter-rows (Table 3).

4. Discussion

4.1. CT-measured number of pores

The number of CT-measured pores for the NC treatment was similar to values found by Udawatta and Anderson (2008) and Udawatta et al. (2008) for soil under row crop management. The values they found were 18 and 26, compared to 21 for the current study. However, the current study found a higher number of macropores for the NC treatment (13 versus 4) than in the studies by Udawatta and Anderson (2008) and Udawatta et al. (2008).

A greater number of CT-measured pores usually indicate a higher level of macroporosity since most of the detected pores are in the macropore range ($> 1000 \mu\text{m}$). A greater number of pores allow higher rates of water transmission when pores are water-filled or higher rates of gas transmission when pores are air-filled. Another important aspect of pores not measured in this study is the degree of tortuosity (Perret et al., 1999). Tortuosity is the relative length of pores to sample length. Images in the current study were not adjacent, so calculations of this property were not possible.

Table 3

Average bulk density and saturated hydraulic conductivity as influenced by Compacted and Non-Compacted treatments and soil depth zones; average bulk density as influenced by Trafficked and Non-Trafficked inter-rows; and analysis of variance.

	Bulk density (g cm ⁻³)	Saturated hydraulic conductivity (cm hr ⁻¹)		Bulk density (g cm ⁻³)
Treatment means			Treatment means	
Compacted	1.45 ^a	14.6 ^b	Trafficked	1.39 ^a
Non-Compacted	1.34 ^b	47.1 ^a	Non-Trafficked	1.26 ^b
Depth zone means				
0- to 10-cm	1.33 ^b	80.6 ^a		
10- to 20-cm	1.45 ^a	7.44 ^b		
20- to 30-cm	1.40 ^a	4.46 ^b		
Analysis of variance, $P > F$			Analysis of variance, $P > F$	
Treatment	0.03	0.02	Treatment	0.02
Depth	<0.01	<0.01		
Treatment by depth	0.02	<0.01		

The ANOVA table represents significance levels between treatments, among depths, and interactions for treatment by depth.

Within columns, values followed by the same letter for the treatments or depths are not significantly different at the 0.05 probability level.

Table 5

Average CT-measured number of pores (pores, macropores, and coarse mesopores), porosity (porosity, macroporosity, and coarse mesoporosity), area of largest pore, circularity, and fractal dimension of macropores as influenced by Trafficked and Non-Trafficked inter-row treatments and soil scan depths along with the analysis of variance.

	Number			Porosity			Area of largest pore (mm ²)	Circularity	Fractal dimension
	Pores	Macropores	Coarse mesopores	Porosity (m ³ m ⁻³)	Macro-porosity (m ³ m ⁻³)	Coarse mesoporosity (m ³ m ⁻³)			
Treatment means									
Trafficked	10 ^b	6 ^b	4 ^b	0.0156 ^b	0.0153 ^b	0.0003 ^b	36.6 ^a	0.77 ^a	1.19 ^a
Non-Trafficked	35 ^a	22 ^a	13 ^a	0.040 ^a	0.039 ^a	0.0010 ^a	36.2 ^a	0.80 ^a	1.29 ^a
Scan depth means									
3.0 cm	22 ^b	14 ^b	8 ^{a,b}	0.028 ^{a,b}	0.027 ^{a,b}	0.0006 ^{a,b}	35.3 ^a	0.78 ^a	1.24 ^{a,b}
4.5 cm	29 ^a	19 ^a	10 ^a	0.040 ^a	0.039 ^a	0.0006 ^{a,b}	37.2 ^a	0.79 ^a	1.32 ^a
6.0 cm	26 ^{a,b}	14 ^b	12 ^a	0.028 ^{a,b}	0.027 ^{a,b}	0.0009 ^a	43.2 ^a	0.78 ^a	1.27 ^{a,b}
7.5 cm	12 ^c	7 ^c	5 ^b	0.016 ^b	0.016 ^b	0.0004 ^b	29.9 ^a	0.80 ^a	1.13 ^b
Analysis of Variance, <i>P</i> > <i>F</i>									
Treatment	<0.01	<0.01	0.02	<0.01	<0.01	0.02	0.97	0.72	0.12
Depth	<0.01	<0.01	0.03	0.10	0.10	0.12	0.94	0.99	0.13
Treatment by depth	0.01	<0.01	0.11	0.17	0.17	0.44	0.38	0.88	0.69

The ANOVA table represents significance levels among treatments and depths for the measured parameters. Within columns, values followed by the same letter for the treatments or the depths are not significantly different at the 0.05 probability level.

4.2. CT-measured porosity

CT-measured porosity values for row crop treatments found by Udawatta and Anderson (2008) and Udawatta et al. (2008) were lower than those of the NC treatment in the current study (0.006 and 0.007 versus 0.020) but were similar to the C treatment (0.0071) of the current study. This may be due in part to averaging over deeper depths in the experiments by Udawatta and Anderson (2008) and Udawatta et al. (2008). Averaging over deeper depths which have lower porosity will reduce the treatment mean.

CT-measured porosity is highly correlated with macroporosity since most of the porosity was measured in the macropore range. Water flow is highly correlated with macroporosity for these soils when saturated as illustrated with the high correlation of K_{sat} with macroporosity ($r=0.79$, Table 4).

4.3. Fractal dimension of macroporosity

Fractal dimension of macropores indicates information regarding the space filling nature of macropores. Larger fractal dimension values indicate that the macropores are more space filling in the image. Fractal dimension values for the NC treatment (1.17) were between values determined for a row crop treatment (1.06 and 1.29) measured by Udawatta and Anderson (2008) and Udawatta et al. (2008). Rachman et al. (2005) found slightly higher fractal dimension values of 1.31 for row crop management and 1.12 for a deposition zone compared to the NC (1.17) and C (1.11) treatments of the current study.

Decreasing fractal dimension values as a function of soil depth as found in the current study has also occurred in other work (Rachman et al., 2005). A decrease of 13% was found for the fractal dimension from the first to third depths of the current study, whereas Rachman et al. (2005) found a decrease of 17% from the 5 to 15 cm depth. Udawatta and Anderson (2008) reported a 19% decrease in fractal dimension from the 5 to 45 cm depth in claypan soils.

4.4. Saturated hydraulic conductivity

Saturated hydraulic conductivity is often related to CT-measured pore parameters as was found in this study. Rachman et al. (2005) showed that CT-measured macroporosity was positively related with K_{sat} ($r=0.95$). In this study, CT-measured macroporosity was the single parameter most correlated with K_{sat} ; the experiment by Udawatta and Anderson (2008) found fractal dimension of CT-measured macroporosity to be the parameter most correlated with

K_{sat} ($r=0.87$). Developing the best two-parameter regression equation, Udawatta and Anderson (2008) found bulk density as well as fractal dimension to be significant ($R^2=0.76$). Udawatta et al. (2008) found the best two-parameter relationship included number of macropores and bulk density, but only 45% of the variability in K_{sat} was explained by these properties.

These studies only evaluated the two-dimensional CT parameters from images and did not assess tortuosity of pores which is also useful in refining predictive models (Perret et al., 1999). Future efforts to develop a better pore-scale model will improve estimation of water flow in soils. However, higher resolution scanning and image analysis costs needed for these studies often restrict analyses to a small number of sample cores and limit the assessment of variations among land treatments which affect saturated hydraulic conductivity.

5. Conclusions

This study examined the changes of pore parameters measured using X-ray computed tomography as affected by a compacted treatment compared to a non-compacted soil. The moderate compaction treatment used for this study reduced number of pores and porosity within the upper 10 cm of the soil. The C treatment increased bulk density by 8% and decreased CT-measured number of pores by 71%, number of macropores by 69%, and coarse mesopores by 75%. The C treatment was also found to significantly decrease CT-measured porosity and macroporosity by 64%. Differences between treatments for the parameters were most pronounced in the upper 10 cm; differences between treatments were not significant below 20 cm. Soil saturated hydraulic conductivity was also significantly reduced by compaction. The CT-measured pore parameters (macroporosity, area of largest pore, and porosity) explained nearly 80% of variability in hydraulic conductivity.

Due to the wet conditions created by the subsurface claypan horizon for these claypan soils, compaction effects may be more enhanced with traffic during the spring planting season. Because perched water tables often occur in these soils, soil profiles are wetter and have less strength to resist compaction. Maintaining macropores in these soils is also critical in order for managers to minimize the enhanced runoff often experienced due to the claypan horizon within the profile of these soils.

The C treatment had a significant impact on soil pores, decreasing their number and relative proportion. The pore fractions most affected by compaction include macropores and coarse mesopores, which are vitally important for ventilation and water transport. Efforts should be made to minimize soil compaction due to its harmful effects on soil

pores, effects which inhibit plant root growth and increase runoff of water and nutrients.

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